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Characterization of NO_x, N_2O, and CO for Lean-Premixed Combustion in a High-Pressure Jet-Stirred Reactor

A high-pressure jet-stirred reactor (HP-JSR) has been built and applied to the study of NO_x and N_2O formation and CO oxidation in lean-premixed (LPM) combustion. The measurements obtained with the HP-JSR provide information on how NO_x forms in lean-premixed, high-intensity combustion, and provide comparison to NO_x data published recently for practical LPM combustors. The HP-JSR results indicate that the NO_x yield is significantly influenced by the rate of relaxation of super-equilibrium concentrations of the O-atom. Also indicated by the HP-JSR results are characteristic NO_x formation rates. Two computational models are used to simulate the HP-JSR and to provide comparison to the measurements. The first is a chemical reactor model (CRM) consisting of two perfectly stirred reactors (PSRs) placed in series. The second is a stirred reactor model with finite rate macromixing (i.e., recirculation) and micromixing. The micromixing is treated by either coalescence-dispersion (CD) or interaction by exchange with the mean (IEM) theory. Additionally, a model based on one-dimensional gas dynamics with chemical reaction is used to assess chemical conversions within the gas sample probe.

Introduction

This is the first study to use a high-pressure jet-stirred reactor (HP-JSR) to examine pollutant formation. The present HP-JSR has been designed for 15 atm, although in the present work the maximum pressure treated is 7.1 atm. The pressure range of 1 to 7 atm, though less than the 10 to 15 atm range of heavy-duty combustion turbines, is useful, since most of the sensitivity in the relaxation of superequilibrium O-atom, and thus NO_x formation, lies in the 1 to 7 atm range.

Because the jet-stirred reactor (JSR) is useful for gaining information on chemical effects for high-intensity combustion, it was decided to extend its operation from the usual pressure of 1 atm to high pressure. In this regard, it is worth recalling that the JSR has assisted in the understanding of gas turbine combustion since the development of the first JSR in the 1950s. Early on, the JSR was used to study flameout and global oxidation rates, and since the 1970s, the JSR has been used to study pollutant formation. The JSR is advantageous for the present work because it simulates the pollutant chemistry of LPM combustion turbines. In LPM combustors, the NO_x mainly forms in the shear layers and adjacent recirculation zones, which are regions of superequilibrium free radical concentrations. The JSR contains the following features: its flow-field consists of one or more strong jets (i.e., shear layers) and one or more recirculation zones, and it is filled with superequilibrium free radical concentrations.

The challenge facing the JSR designer is attainment of the well-stirred reactor (WSR) condition, which is defined as spatial uniformity in the time-mean temperature and species concentrations. The WSR condition requires that the characteristic chemical time of the event being examined, in this case NO_x formation, be long compared to the characteristic mixing times in the JSR. There is substantial evidence that the WSR condition is essentially satisfied for NO_x formation in the 1 atm JSR operated lean.

The WSR condition is attained at 1 atm because free radical recombination rates are relatively slow, and, thus, high levels of O, H, and OH concentrations essentially fill the reactor uniformly. This is implied by Fig. 1, due to Nicol (1995). In this figure, the time for the O-atom to relax from its peak concentration (which occurs near the end of the premixed flame zone) to 110 percent of its thermochemical equilibrium concentration is plotted versus pressure for a typical LPM combustion fuel-air ratio. Figure 1 is based on chemical reactor modeling, i.e., a perfectly stirred reactor (PSR) followed by a plug flow reactor (PFR), though essentially equivalent behavior to that shown may be obtained by one-dimensional, laminar flame modeling. The chemical kinetic mechanism used is that of Miller and Bowman (1989). For 1 atm, the relaxation time shown in Fig. 1 is about 20 ms, which is long compared to the mean residence time of 2 to 4 ms of the JSR, and, thus, to the characteristic mixing times of the JSR. Since the NO_x formation in LPM combustion is strongly influenced by the O-atom behavior through both the nitrous oxide and Zeldovich mechanisms (Nicol et al., 1995), the JSR is essentially a WSR with respect to NO_x formation. For pressures in the vicinity of 5 atm and greater, however, Fig. 1 shows that the O-atom relaxation time is short, about 1 to 2 ms. Thus, a high pressure JSR operated lean at 2 to 4 ms, as the case here, cannot be a WSR with respect to NO_x formation. In particular, the characteristic chemical time is not long compared to the mixing time of the large recirculation eddy. In order to attain the WSR condition, the HP-JSR must be operated at a residence time of 0.5 ms or less. (A 0.5 ms HP-JSR is under development.)

Attention is focused on the O-atom because it has a greater impact on NO_x formation than the other major free radicals in flames, i.e., H-atom and OH (Nicol, 1995). For LPM combustion, the O-atom accounts for at least about 50 percent of the NO_x formed by the nitrous oxide pathway (through the reaction

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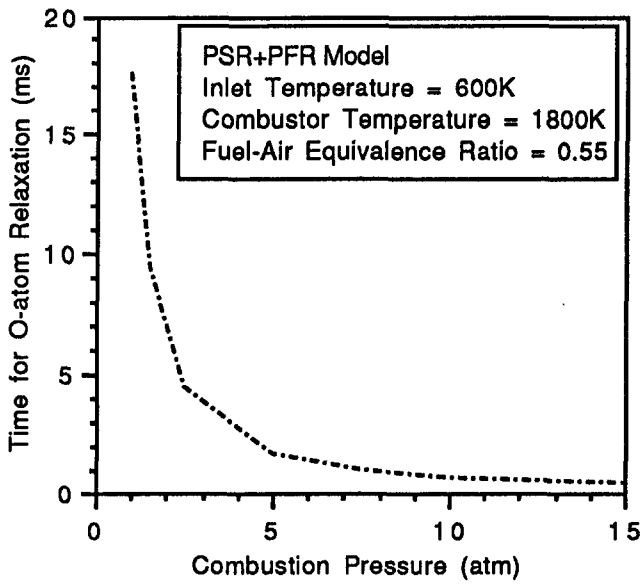

Fig. 1 O-atom relaxation time versus pressure (Nicol et al., 1995)

$N_2O + O \rightarrow 2NO$), and the O-atom accounts essentially for all of the NO_x formed by the Zeldovich pathway (through the reaction $N_2 + O \rightarrow NO + N$, followed by oxidation of the N-atom to NO). The H-atom accounts for the balance of the NO_x formed by the nitrous oxide pathway (through the reaction $N_2O + H \rightarrow NO + NH$, followed by oxidation of the NH radical to NO). Also, the H-atom destroys N_2O, and thus, influences the N_2O concentration in the combustion reactor (through the reaction $N_2O + H \rightarrow N_2 + OH$). However, since the H-atom is of less importance overall than the O-atom in establishing NO_x formation in LPM combustion and since the trend of H-atom relaxation is generally similar to that of the O-atom, the H-atom relaxation is not explicitly treated herein.

The other free radical of significance to NO_x formation is CH. Modeling indicates that the CH radical has a very short lifetime (of order 0.1 ms) in LPM combustion (Nicol, 1995), is confined to the flame zone, and does not survive into post-flame zones (as occurs for O, H, and OH). Although there is evidence for the formation of prompt NO_x in LPM combustion (by the reaction $CH + N_2 \rightarrow HCN + N$, followed by oxidation of the HCN and N-atom to NO), the short lifetime of the CH radical and chemical reactor modeling (Nicol et al., 1995) indicate that prompt NO_x is not a major source of NO_x in LPM combustion, unless the fuel-air equivalence ratio is greater than about 0.65, or the residence time of the combustion reactor is very short. For the present study, the prompt mechanism is regarded as a secondary source of NO_x, the nitrous oxide and Zeldovich mechanism are assumed to account for most of the NO_x, and the relaxation time of the O-atom is assumed to define the characteristic time for NO_x formation under the influence of high concentrations of free radicals.

The non-WSR nature of the HP-JSR has provided impetus for the development and application of two models. The first is a chemical reactor model (CRM) in which two perfectly stirred reactors (PSRs) placed in series are used to simulate the reactor. The second model is a stirred reactor with finite rate macromixing (i.e., recirculation) and micromixing. The micromixing is treated by either coalescence-dispersion (CD) or interaction by exchange with the mean (IEM) theory. In addition, a probe model based on one-dimensional gas dynamics with chemical reaction is used to assess probe chemical conversions.

In the following sections of this paper, the HP-JSR system is described; the experimental results for the combustion of methane are presented, discussed, and compared to the literature; and the computational models are described briefly and compared to the measurements. Results are given and discussed for NO_x, N_2O, and CO. The paper provides an understanding of how NO_x forms in high-pressure, jet-stirred combustion, and it provides a useful comparison to other laboratory results, including results for practical LPM combustors.

Experimental Setup and Approach

The experimental facility and high-pressure, jet-stirred reactor (HP-JSR) are described by Steele (1995). The facility consists of the pressure vessel (i.e., cylinder) in which the HP-JSR, its premixing chamber, and its electrical air heater are located, and it consists of the equipment for the distribution, metering, and control of the air and fuel. Figure 2 shows the pressure vessel with the HP-JSR and the premixing chamber. The pressure vessel also contains ports (not shown in Fig. 2) for the insertion of the thermocouple and gas sampling probes through the side wall of the pressure vessel into the HP-JSR. The dimensions of the components shown in Fig. 2 are as follows: pressure vessel (stainless steel): height is 1 m (nominal), inside diameter is 127 mm, and wall thickness is 9.5 mm; premixing chamber (stainless steel): height is 240 mm, inside diameter is 25 mm, distance from point of fuel injection holes to nozzle block is 115 mm, and diameter of each of the 4 fuel injection holes is 1 mm; jet-stirred reactor (Zirconia): outside diameter is 64 mm, outside height is 51 mm, inside diameter at widest point is 12.7 mm, and inside height is 22.2 mm; and nozzle block (Inconel): height is 9.5 mm, and diameter of jet is 1 mm.

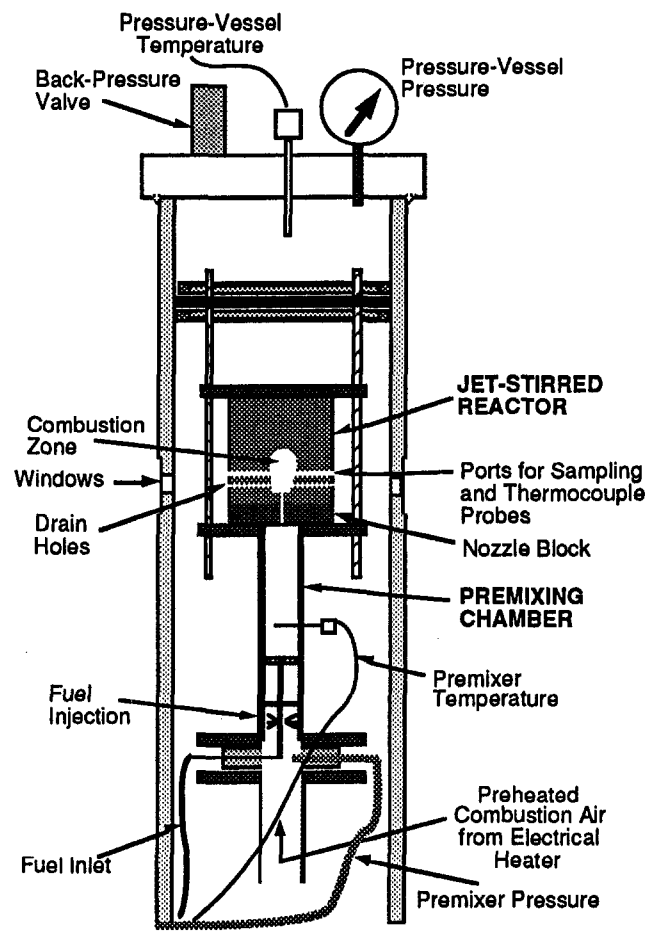

Fig. 2 Schematic of HP-JSR with premixing chamber

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Air from a mechanical compressor, of maximum pressure 7.5 atm, is filtered and forced to flow through the pressure vessel. This air dilutes and carries away the effluent of the HP-JSR. The pressure level of the pressure vessel is controlled by a back-pressure valve located at the exit of the pressure vessel. Although the pressure within the HP-JSR is slightly greater than the pressure of the air in the pressure vessel, the nominal pressure of the HP-JSR, which is the pressure reported below, is taken to equal the pressure of the air in the pressure vessel. The pressure vessel and other components are designed for a pressure of 15 atm, though for the present study, because of the limitation of the mechanical compressor used, the maximum pressure tested is 7.1 atm.

The combustion air for the HP-JSR is provided either from bottled air or from a separate stream from the mechanical compressor. This air is preheated and injected into the premixing chamber (see Fig. 2). This chamber serves the following two functions: (1) the gaseous fuel (from bottles) and the combustion air are premixed; and (2) the high pressure required to create the high velocity jet of the HP-JSR is attained. For experiments with a HP-JSR pressure of 6.5 atm or less and a residence time of 4 ms or greater, the mechanical compressor provides the combustion air. For experiments of greater pressure levels and/or shorter residence times, the pressure of the mechanical compressor is inadequate, and bottled air is used as the combustion air. The bottled air is tested for traces of N_2O, and only bottles with negligible levels of N_2O are used. The fuel/air mixture entering the HP-JSR is assumed to be perfectly premixed. Several features of the design support this assumption: (1) the high pressure drop and small size of the jets which inject the fuel into the premixing chamber; (2) the baffle plates used in the premixing chamber; (3) the residence time in the premixer is long (i.e., at least 0.2 s); and (4) the small size of the premixed fuel/air jet entering the HP-JSR.

The shape of the HP-JSR chamber is identical to that shown by Steele et al. (1995) for the 1 atm JSR. The HP-JSR is stirred by the centered, high velocity jet of 1 mm diameter. The Mach number of the inlet jet is nominally sonic for 2 ms operation, and nominally 0.5 for 4 ms operation. The jet is directed upwards, entrains recirculating gas, and is reflected at the reactor's back wall, thereby creating a strong downwards recirculation. The flow pattern within the HP-JSR is identical to that shown by Steele et al. (1995) for the 1 atm JSR with a single, centered jet. The volume of the HP-JSR is 2 cc; that is, all dimensions are one-half the dimensions of the 1 atm, 16 cc JSR used by Steele et al. (1995). Gases exit the reactor through 6 holes, including 4 holes (each of diameter 2.4 mm) at the bottom of the reactor, and 2 holes (each of diameter 3.2 mm) at 64 percent elevation, which is the same elevation as the holes for the thermocouple and sampling probes. The bottom holes encourage the gas to fully recirculate. Application of jet entrainment formulas (Beér and Chigier, 1983) shows that the average fluid particle makes about 4 to 6 cycles of the reactor. (A reactor nozzle with small, multiple jets would permit a greater number of cycles. Such a nozzle was recently built for the HP-JSR. For the 1 atm JSR, Steele et al. (1995) reported little difference in the NO_x yield of the single and multiple jet configurations.)

The small (2 cc) volume minimizes the size and cost of the high pressure rig, and the cost of the fuel and air. No detriment has been found from this approach. Because of the high loading at high pressure, the HP-JSR operates nearly adiabatically. Measurements for 1 atm show that the 2 cc JSR gives the same NO_x as a function of combustion temperature as JSRs tenfold to a hundred-fold larger (see Steele et al., 1995).

Combustion temperature is measured by a small type-R thermocouple. The thermocouple bead is coated with ceramic material to prevent catalytic reaction, giving a bead diameter of about 1 mm, and the measured temperature is corrected for heat loss by calculation. The correction is small (i.e., about 30 to 40°C) because the HP-JSR chamber approximates a black-body cavity. Unless stated otherwise below, the temperature reported is that measured (and corrected) for the recirculation zone of the HP-JSR. In this zone the temperature is uniform, and has attained its peak value (see Fig. 3), which is close to that for adiabatic equilibrium.

Gas sampling is conducted with a microprobe. Two probes have been developed; one a quartz tube (Steele, 1995), the other a ceramic tube (Horning, 1996). With each, the inlet of the tube is closed to a very small diameter so that the inlet flow is choked, and a small segment of supersonic flow is obtained before the flow undergoes shock recompression to the back pressure of about 0.3 atm imposed by the sampling pump. External cooling of the probe is obtained by permitting a small amount of the pressure-vessel air to escape and flow rapidly

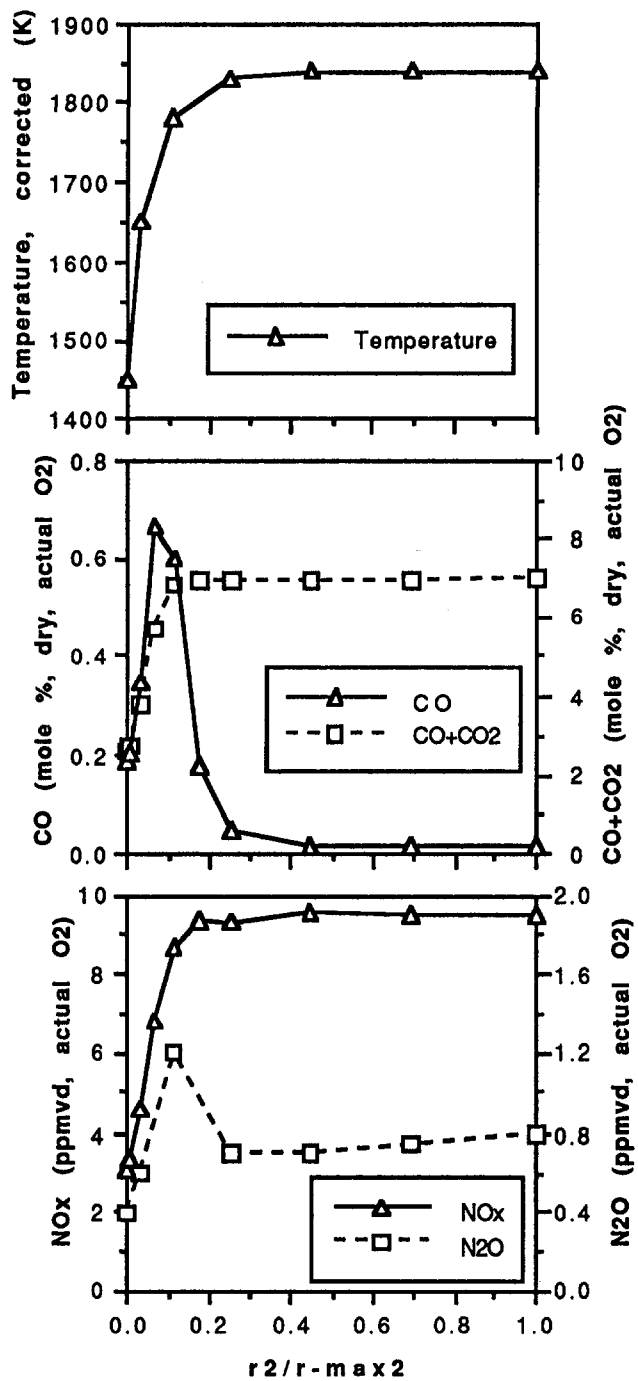

Fig. 3 Typical HP-JSR profiles from centerline to the inside wall

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over the probe as it passes through the port in the side wall of the pressure vessel. The useful lifetime of a quartz probe is about 8 hours. The failure mode of the quartz probe is devitrification, which is accelerated by high pressure. In order to obtain improved lifetime, a ceramic probe has been developed (Horning, 1996). Both probes have undergone extensive modeling with the probe model. Unless stated otherwise, the species data given below are for the recirculation zone of the reactor, and were obtained with the quartz probe. For sampling from the recirculation zone of the HP-JSR (with CO concentration of order 100 ppmv), the modeling indicates that chemical conversions within the probe are small. Essentially, NO_x is conserved, though conversion of NO to NO_2 occurs. Loss of N_2O is less than 10 percent (relative), and CO oxidation is 10 to 20 percent (relative).

The gas sample line connecting the sample probe to the gas analyzers is wrapped with heating tape and maintained at a temperature sufficient to prevent condensation and loss of NO_2 by absorption into water. The water is removed in ice cooled impingers, while maintaining minimum gas (e.g., NO_2) contact with the water. Sampling with newly dried impingers gives identical results to sampling with wet impingers, indicating a negligible loss of NO_2 in this work. Analysis is by process gas analyzers (chemiluminescence for NO_x, NDIR for CO and CO_2) and by electron capture gas chromatography for N_2O. The uncertainty estimated for the NO_x measurements is ±20 percent (relative).

Experimental Results and Discussion

The measurements reported herein are for methane burned over the combustion temperature range of 1550 to 1850 K and the pressure range of 1 to 7.1 atm. The combustion temperature was varied by varying the fuel-air ratio (for the particular inlet temperature). The 1 atm JSR results for methane fuel, though reported previously by Steele et al. (1995), are summarized below in order to provide completeness to the present paper and a benchmark for comparison of the HP-JSR results. The complete experimental results are contained in the theses of Steele (1995) and Horning (1996).

Summary of Results on NO_x Formation for 1 atm Combustion of Methane. The NO_x data obtained for 1 atm methane combustion cover the fuel-air equivalence (ϕ) range 0.46 to 0.7, the combustion temperature range 1550 to 1850 K, and the residence time range 1.6 to 7.2 ms, for inlet temperatures of 300 and 600 K (nominal) and for both single and multiple jet configurations (Steele, 1995; Steele et al., 1995). The NO_x data vary exponentially with combustion temperature and linearly with residence time, and they agree well with the following best fit expression:

$$NO_x(\text{ppmvw, actual } O_2)/\tau(\text{ms}) = 2.29 \times 10^6 \exp(-24290/T(K)), \quad (1)$$

where τ(ms) is the JSR residence time in milliseconds, $T(K)$ is measured (corrected) combustion temperature in degrees Kelvin, and the NO_x is expressed as parts per million by volume, wet basis (ppmvw), assuming no O_2 correction.

The 1 atm NO_x data are very well modeled by assuming the JSR to be a PSR at the experimental temperature and residence time (Steele, 1995). This finding, the linear dependency of the NO_x on residence time shown by Eq. (1), and the high levels of CO observed throughout the JSR (Steele, 1995) give substantial evidence for the superequilibrium free radicals (O, OH, and H) filling the JSR and affecting an essentially uniform NO_x formation rate. That is, the whole JSR is a flame zone. Thus, Eq. (1) may be taken as the NO_x formation rate for 1 atm high-intensity, lean-premixed, methane-air flame zones. In applying Eq. (1), the user must decide the residence time of the flame zone (i.e., the time over which high free radical concentrations persist). This is either the combustor residence time (as the case with the 1 atm JSR) or the O-atom relaxation time (see Fig. 1), whichever is smaller.

If only the 600 K inlet temperature data are curve fit, Eq. (1) becomes $NO_x/\tau = 1.28 \times 10^7 \exp(-27230/T)$. The comparison of this expression with Eq. (1) indicates that the overall NO_x activation energy increases slightly with increasing inlet temperature. That is, for the highest combustion temperatures, the NO_x concentration is slightly higher for the 600 K inlet temperature than for 300 K. This tendency is also noted in PSR modeling. However, for a 1 atm JSR with a large inlet jet (Williams, 1995), and thus relatively slow jet eddy mixing, the NO_x is found to decrease with increasing inlet temperature (for fixed combustion temperature). This behavior may be a response of the NO_x chemistry to the decreased fuel/air ratio permitted by the higher inlet temperature. Adjustment of the 600 K data to a dry, 15 percent O_2 basis changes Eq. (1) to $NO_x/\tau = 4.22 \times 10^5 \exp(-22115/T)$.

Reactor Uniformity. Representative profiles of temperature, CO, CO + CO_2, NO_x, and N_2O for the HP-JSR are shown in Fig. 3. These measurements were obtained by Horning (1996) for the case of 6.5 atm pressure, 4 ms residence time (nominal), and 300 K inlet temperature (nominal), using the ceramic sampling probe. The corrected temperature for the recirculation zone is 1840 K; the fuel-air equivalence ratio is 0.7. The profiles extend from the centerline (0) to the inside wall (1.0) of the HP-JSR, and indicate two-zone behavior. Except for the jet (centerline) zone, the HP-JSR is uniform in temperature and NO_x. The outer 70 percent of the HP-JSR, on the r^2 basis shown, is uniform in all quantities. This is the recirculation zone.

The CO profile is indicative of the two-zone behavior. The CO forms in the jet, reaches peak concentration in the jet boundary layer, and then oxidizes and reaches a low (though greater than equilibrium) concentration in the recirculation zone. Since O-atom and CO concentrations are well correlated in LPM combustion (Nicol, 1995), the CO profile may be taken as an indicator of the O-atom behavior and the zones of high and low free radical concentrations. By this reasoning, the highest rate of NO_x formation occurs mainly in the jet boundary layer. This situation is different from that argued above for the 1 atm JSR, which is NO_x formation dispersed throughout the reactor and recirculation zone.

The recirculation zone temperature of Fig. 3 is relatively high, 1840 K. Profile measurements (not shown here) have also been obtained for lower combustion temperatures (Steele, 1995; Horning, 1996). For these experiments, the profile trends for temperature, NO_x, and CO + CO_2 are similar to the profiles in Fig. 3. However, the CO behavior in the outer part of the lower-temperature reactor changes from the flat profile of Fig. 3. Particularly, the CO close to wall increases due to the flow of less-reacted gas from the jet zone into the outer region of the recirculation zone. At lowest temperatures (less than 1650 K) and shortest residence times (2 ms nominal), high CO concentrations (of magnitude 3000 ppmv) fill the recirculation zone. In this case, the dispersion of super-equilibrium CO throughout the HP-JSR is similar to the behavior observed in the 1 atm JSR. Also, at the lower combustion temperatures, the N_2O does not relax from its peak concentration but remains at elevated concentration throughout recirculation zone (Horning, 1996).

Probe modeling of sampling from regions of high CO concentration (and thus, high free radical concentrations) shows that the CO conversion to CO_2 can be as great as 60 percent (relative), compared to 10 to 20 percent conversion stated above for sampling from the recirculation zone (with CO concentration of order 100 ppmv). Thus, the high CO measurements should be treated qualitatively rather than quantitatively. The NO_x and

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N_2O conversions remain negligible to small, as stated above for the low-CO recirculation zone.

Results for High Pressure Combustion of Methane. The measurements for the HP-JSR cover the pressure range 1 to 7.1 atm and the combustion temperature range 1550 to 1850 K, and have residence times of 2 and 4 ms nominal (1.6 to 4.5 ms actual) and inlet temperatures of 300 and 600 K nominal (297 to 605 K actual). Figure 4 shows the complete dataset for the NO_x and N_2O concentrations for the 1 to 7.1 pressure range from the work of Steele (1995). About one-half of the data for 1.0 atm were obtained in the 16 cc (1 atm) JSR. All other data were obtained in the 2 cc HP-JSR. (As shown by Steele et al. (1995) JSR size does not affect NO_x.) At the top of the band of the NO_x data, most of the points are for 1 atm; whereas, at the bottom of the band most of the points are for 7.1 atm. That is, the NO_x exhibits a slightly negative dependency on pressure. This tendency is also noted in the data of Joshi et al. (1994). Analysis of the NO_x data in Fig. 4, assuming an NO_x pressure (P) dependency of

$$NO_x \approx P^n \tag{2}$$

gives $n = -0.20$ as the average value, and $n = -0.10$ to -0.35 as the range.

The 1 atm data plotted in Fig. 4 are subject to the linear increase in NO_x with residence time expressed by Eq. (1). However, the high pressure (4.7 and 7.1 atm) data are nearly neutral with respect to residence time; the doubling of the residence time from 2 to 4 ms (nominal) causes an increase of 0 to 20 percent (relative) in the NO_x. This nearly neutral result may be explained by the O-atom behavior discussed above. High concentrations of free radicals no longer fill the whole reactor, and thus, the free radical relaxation time rather than reactor residence time affects the NO_x yield. Although the rate of NO_x formation in the high free radical region of the HP-JSR is about five-fold greater than for the 1 atm JSR, this effect is neutralized by the fast relaxation of the free radicals at high pressure.

For high free radical zones of 1800 K temperature, the rate of NO_x formation is about 2 ppmvd (15 percent O_2) per millisecond at 1 atm (based on Eq. (1)), and about 10 ppmvd (15 percent O_2) per millisecond at 7 atm (based on chemical reactor modeling by Nicol (1995) and Horning (1996).

With respect to inlet temperature, the HP-JSR NO_x data show little to no sensitivity to the doubling in inlet temperature from 300 to 600 K (nominal).

The N_2O data plotted in Fig. 4 show a greater scatter than the NO_x data, and indicate three regimes of behavior: 1 atm, high pressure/low temperature, and high pressure/high temperature. The 1 atm data show a slight decrease with increasing temperature; also, the highest data are for 2 ms residence time, the lowest for 4 ms. The 4.7 and 7.1 atm N_2O data are 3 to 4 fold greater than the 1 atm data, except for the high temperature regime. When the combustion temperature is high (i.e., about 1800 K and greater) and the residence time is sufficient (i.e., 4 ms), the N_2O has an opportunity to relax towards its equilibrium concentration. High pressure promotes this effect; the data points (filled triangles) at 1850 K and 0.6 to 0.7 ppmvw are for the case of 7.1 atm and 4 ms. For 2 ms residence time, however, the N_2O remains in the 2 to 3 ppmvw range for all temperatures, since the time is inadequate for relaxation towards equilibrium.

Although much of the NO_x forms from oxidation of the N_2O, especially for combustion temperatures below 1800 K (see Nicol et al., 1995), a correlation between the NO_x and N_2O is not evident from Fig. 4. This is because the formation of NO_x from N_2O also depends on the O-atoms (and H-atoms).

Comparison to LPM Combustion Turbine Data. In Fig. 5, the NO_x data from Fig. 4 are adjusted to a dry, 15 percent O_2 basis and compared to best-fit results from the recent gas turbine literature. (The best-fit curves are ours.) The jet-stirred reactor data are bracketed by those data from the literature which either have a nil to weak pressure dependency or are for an engine premixer/combustor run at 1 atm. This includes the following: (1) the data of Leonard and Stegmaier (1993) obtained for a porous-plate laboratory combustor, for which insignificant dependency of the NO_x on pressure, inlet temperature, and overall residence time was reported; (2) the data of Joshi et al. (1994) obtained using prototype premixers and combustor segments, which indicate a slightly negative pressure exponent (Eq. (2)) similar to that stated above for the HP-JSR data; and (3) the data of Sattelmayer et al. (1990) for the double-cone combustor run at 1 atm. The data of Snyder et al. (1994) for the high penetration premixer also lie relatively close to the jet-stirred reactor data. For certain regimes, the Snyder et al. data show a weak, though generally somewhat positive pressure dependency of the NO_x. The tendency of the NO_x emissions of the practical premixers/combustors (Joshi et al., Sattelmayer et al., Snyder et al.) to lie a factor 1 to 2 above the laboratory data (of this study and Leonard and Stegmaier, 1993) is attributed to more nearly perfect premixing possible in the laboratory rigs than in the practical hardware. Other data plotted are from Snyder et al. (1994) for the low penetration premixer, Aigner et

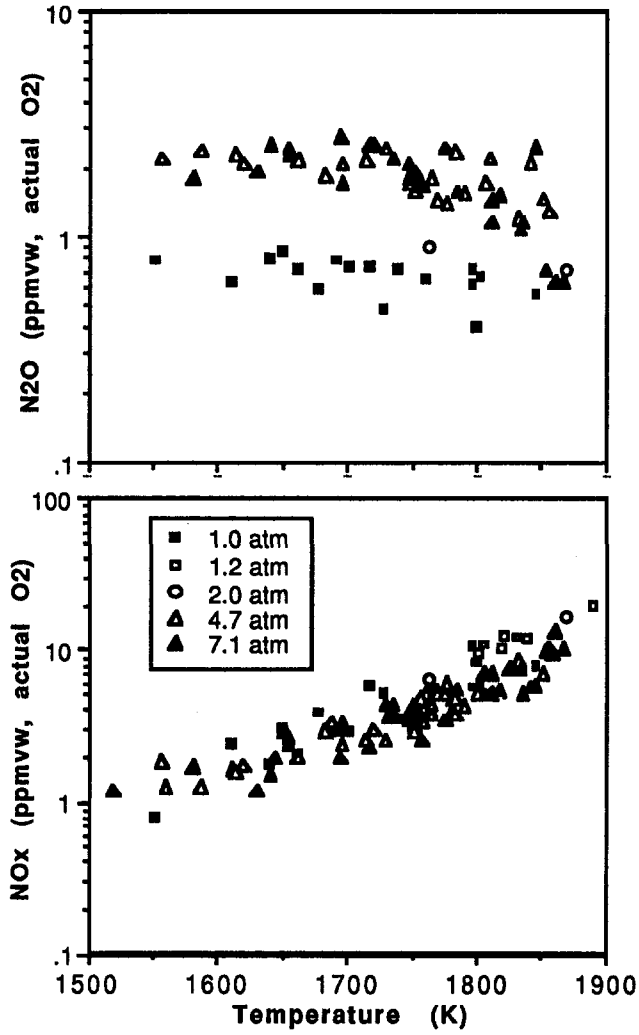

Fig. 4 Complete NO_x and N_2O datasets for methane combustion over the 1 to 7.1 atm pressure range. Residence time = 2 and 4 ms (nominal). Inlet temperature = 300 and 600 K nominal. The 1 atm data are for 2 cc and 16 cc JSRs, using both single and multiple inlet jets. Data for all other pressures are for the 2 cc, single jet HP-JSR.

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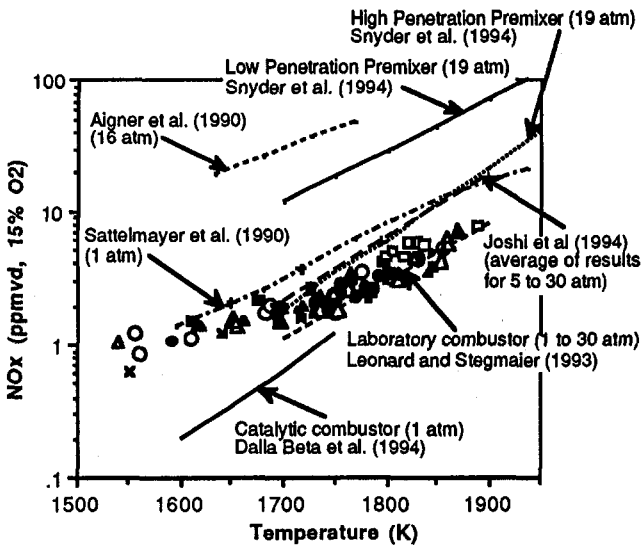

Fig. 5 Comparison of complete NO_x dataset to recently published NO_x data for LPM combustors

al. (1990) for the double-cone burner, and Dalla Betta et al. (1994) for a laboratory catalytic combustor.

For the complete jet-stirred reactor database, over the temperature range of 1700 to 1900 K, the Arrhenius-type best fit is

$$NO_x(\text{ppmvd, 15 percent } O_2) = 1.72 \times 10^6 \exp(-23650/T(K)), \quad (3)$$

corresponding to an overall activation energy of NO_x formation of 47.0 kcal/gmol. The Leonard and Stegmaier data correspond to an overall activation energy of 57 kcal/gmol. Other results plotted in Fig. 5 give overall activation energies in the range from 35 to 70 kcal/gmol.

Chemical reactor modeling has been used by Nicol (1995) to determine NO_x overall activation energies for the 1700 to 1900 K range. The combustor is simulated by a PSR (of either incipient-blowout or 2 ms residence time) followed by a PFR (plug flow reactor). Total residence time is 6 ms. The results given below in Table 1 show that the NO_x activation energy is sensitive to the chemical kinetic mechanism, the pressure level, and the residence time split. The greater activation energies for the blowout PSR are due to the relatively long residence time of the PFR, and, thus, the opportunity for post-flame, thermal NO_x formation. Thermal NO_x, i.e., NO_x formation due to the Zeldovich mechanism under the influence of equilibrium O-atom, has an activation energy of 135 kcal/gmol. If thermal NO_x is eliminated from the computation, and only the NO_x due to superequilibrium free-radical attack on the N_2 (in the Zeldovich, nitrous oxide, and prompt mechanisms) is considered, the range of the activation energies for the full range of conditions of Table 1 is narrowed; 47.7 to 55.5 kcal/gmol by the MB mechanism (indicating little pressure sensitivity), and 54.2 to 77.6 by the DB mechanism.

Table 1 Overall activation energies of NO_x formation

PSR Residence Time (ms)	Pressure Range	
	1 to 10atm	10 to 30atm
	Miller-Bowman (1989) Mechanism (MB)	
blowout	56.8	71.0
2.0	48.8	60.4
	Drake-Blint (1991) Mechanism (DB)	
blowout	65.1	90.5
2.0	56.6	83.8

Chemical Reactor Modeling of the HP-JSR

Chemical reactor modeling for the HP-JSR is based on two PSRs in series (Steele, 1995). This configuration is used in order to simulate the two-zone behavior implied by the O-atom relaxation (Fig. 1) and the CO profile (Fig. 3). The first PSR is used to provide a relatively small zone of high free-radical concentrations and high NO_x formation rate. The second PSR is used to simulate the significantly reduced free-radical activity of the relatively large recirculation zone.

The ratio of volumes of the two PSRs is selected so that the minimum CO concentration measured in the recirculation zone is matched to the modeled result for the second PSR. The temperature is matched also. The volume of the two PSRs is set equal to the volume of the HP-JSR. The CO loss in the probe is modeled and is included as part of the matching.

Alternatives to the two-PSRs-in-series configuration have been evaluated for the HP-JSR (Steele, 1995; Horning, 1996). The replacement of the second PSR with a PFR permits too much CO oxidation, and, thus, gives too low of CO concentration exiting the reactor. This implies that the recirculation zone of the HP-JSR has sufficient mixing to prevent the CO from relaxing to its equilibrium concentration. If the PFR is made

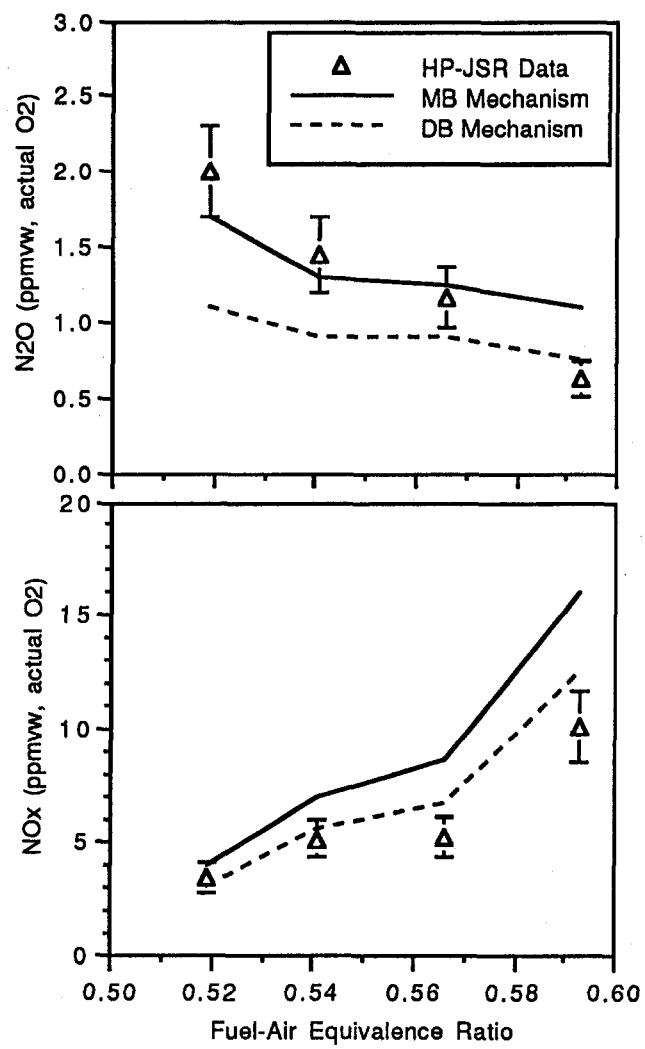

Fig. 6 Comparison of two-PSR model to measurement. Pressure = 7.1 atm, residence time = 3.9 ms, inlet temperature = 605 K. Respective combustion temperatures are 1745, 1810, 1835, and 1865 K.

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small and, consequently, the initial PSR is made large, the minimum CO of the recirculation zone is matched, but the NO_x becomes unrealistically large. (The model of a large PSR and a small PFR does work acceptably for the 1 atm JSR.) The addition of recycle of some of the exit gas of the second PSR into the initial PSR gives too large of CO concentration in the second PSR. On the other hand, if recycle is added to the PSR-followed-by-PFR model, the problem of too low of exit CO returns. The two-PSRs-in-series model works best for methane burned at temperatures below about 1800 K. For higher temperatures, the addition of a small PFR zone after the second PSR gives best agreement to the measured CO (Horning, 1996). The small PFR zone simulates additional relaxation of the CO in the recirculation zone of the HP-JSR. The effect of the small PFR zone on the NO_x prediction is nil.

Results obtained by the two-PSRs-in-series configurations are shown in Fig. 6. The NO_x and N_2O modeled for the second PSR are compared to the data measured for the recirculation zone. Two chemical kinetic mechanisms have been used in this analysis: the Miller and Bowman (1989) mechanism (MB), and the Drake and Blint (1991) mechanism (DB). The trends in the measured NO_x are reproduced by the model. The modeled results obtained with the DB mechanism lie relatively close to the measured NO_x. On the other hand, the measured N_2O is most closely reproduced by the MB mechanism. This implies that the DB mechanism may under predict the NO_x formed via the nitrous oxide pathway, and may compensate for this by over predicting the NO_x formed by the prompt pathway.

Finite Mixing Modeling of the HP-JSR

A reactor model which includes finite rate macromixing and micromixing effects is applied in this section. This numerical model, developed by Tonouchi and Pratt (1995), involves a particle-tracking, control volume approach. The model and its application are described by Tonouchi (1996). The jet and recirculation zones of the HP-JSR are divided into several control volumes having the same axial length. The amount of macromixing within the HP-JSR is modeled using the recirculation ratio R, which is defined as the ratio of the mass flow rate of the jet at the reactor's back wall to the inlet jet mass flow rate.

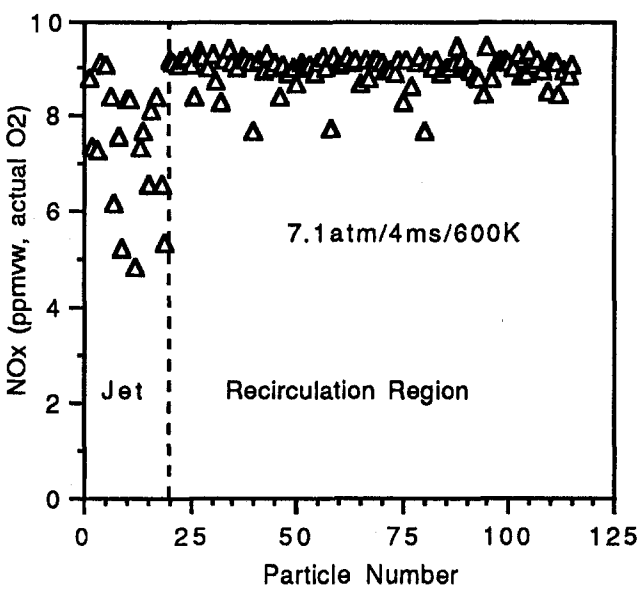

Fig. 7 Application of the finite-rate mixing model to HP-JSR. Pressure = 7.1 atm, residence time = 4 ms, inlet mixture temperature = 600 K, fuel-air equivalence ratio = 0.58. Results are presented as a scatterplot "snapshot" of particle NO_x concentrations. The mean particle NO_x in the jet element is 7.5 ppm. The mean particle NO_x value in the recirculation region is 9.0 ppm.

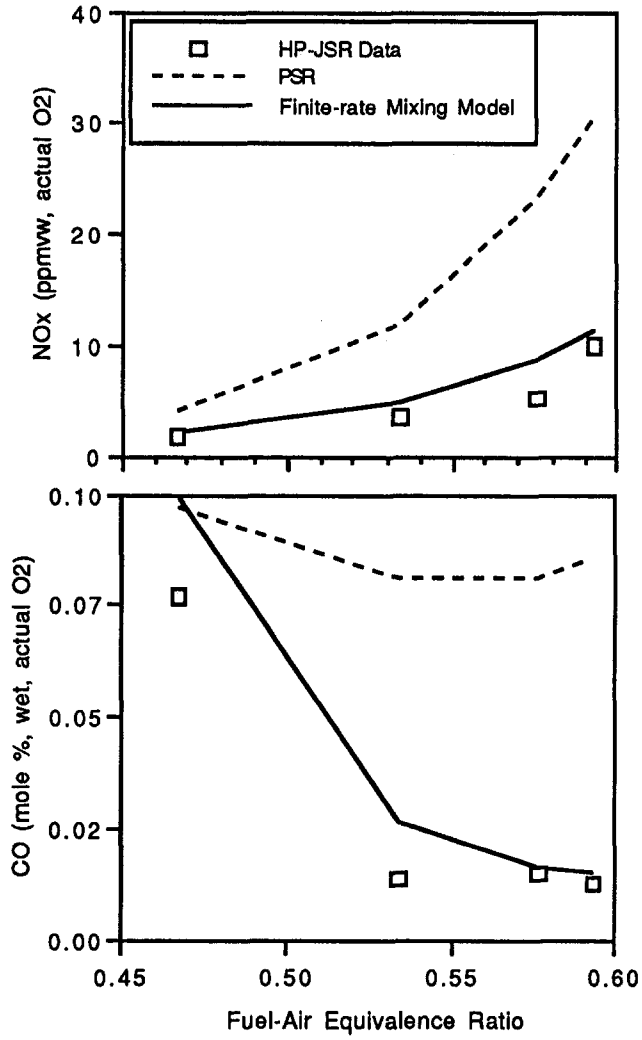

Fig. 8 Comparison of the finite-rate mixing model and the PSR model to the experimental data of Steele (1995). Pressure = 7.1 atm, residence time = 4 ms, and inlet temperature = 600 K.

Finite-rate micromixing within each control volume is modeled using either the CD or IEM theory. In the CD approach, micromixing is simulated by randomly pairing particles to mix (coalesce and disperse). During this CD event, the extensive properties of the paired particles are averaged, while between CD encounters the particles undergo batch reaction. The number of pair encounters for a given time Δt is given by $\beta \Delta t N_{cv}$, where N_{cv} is the number of particles in the control volume, and β is the CD mixing frequency or the reciprocal of the turbulent mixing time. The ratio of β to the feed frequency (i.e., the reciprocal of the mean residence time) is the micromixing parameter I_m. The parameter I_m is prescribed for each control volume by estimating the CD mixing frequency β as the ratio of the local turbulent dissipation rate to the local turbulent kinetic energy. The resulting algebraic expression for I_m requires an estimate of the size of the velocity fluctuations. For the purposes of modeling the experimental data, the rms values of the velocity fluctuations are estimated to be 25 percent of the local mean velocity of the gases flowing in the control volume.

In the IEM theory, the particles' physical properties are determined from an evolution equation that includes the mean properties of the control volume. The IEM algorithm used in this model is similar to the approach used by Correa (1993). In the IEM model, the micromixing parameter is the IEM mixing frequency w. This mixing frequency is related to the CD mixing frequency by a factor of 2 (Villermaux, 1986).

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Figures 7 and 8 show the application of the finite mixing model to the HP-JSR data for 7.1 atm pressure, 600 K inlet temperature, and 4 ms residence time. The chemical kinetic mechanism used is the 24 step/16 species reduced mechanism of Nicol (1995), which is based on the Miller and Bowman (1989) mechanism. The reduced mechanism retains the essential features of methane oxidation and NO_x formation by the Zeldovich and nitrous oxide reactions. The value of R estimated for this application is 6.5, the micromixing is treated by the CD theory, and the reactor is assumed adiabatic. For reference, results obtained by modeling the HP-JSR as a single, adiabatic PSR are also plotted. In this case, the full Miller and Bowman (1989) mechanism is used.

Figure 7 is a scatterplot of NO_x for one instant ("snapshot") for a fuel-air equivalence ratio of 0.58. The scatterplots are for the jet control volume and adjacent recirculation region control volume corresponding to the sample probe path. This figure indicates a low variation of NO_x in the recirculation zone. Figure 8 shows the predictions of the finite-rate mixing model and the PSR model. The finite-rate mixing model results are for the recirculation region control volume corresponding to the sample probe path. The results plotted for NO_x and CO generally show close agreement to the experimental data, and, thus, indicate that the data are influenced by finite-rate mixing effects. Essentially, the same results (not shown) are obtained when the IEM micromixing model is used (Tonouchi, 1996). The results for temperature and N_2O predicted by the finite rate mixing model, though not plotted in Fig. 8, also generally show close agreement to the experimental data. Additionally, although predictions of NO_x by the two-PSRs-in-series model are not shown in Fig. 8, the results in Fig. 8 may be directly compared to the two-PSRs-in-series results of Fig. 6. This comparison may be made because Figs. 6 and 8 pertain to the same experimental conditions (though the measurements were taken on different days).

Conclusions

The high-pressure, jet-stirred reactor (HP-JSR) has been described and shown to exhibit distinct two zone behavior. The HP-JSR provides useful data and insight on NO_x formation for high-intensity, LPM combustion. In particular, the results show how NO_x formation varies as a function of the primary combustion variables, including combustion temperature, pressure, inlet temperature, and residence time, and indicate that the NO_x formation is sensitive to the free radical relaxation rate. Also indicated are characteristic rates of formation of NO_x for LPM flame zones.

Two models have been developed. One is the two-PSRs-in-series model, which conveniently permits evaluation of the HP-JSR, and may be useful generally for the prediction of species concentrations in high-intensity, high-pressure flames. The other is the finite mixing model, which reproduces the HP-JSR species concentrations quite well, and indicates the degree to which finite mixing affects the HP-JSR. The finite mixing model shows promise for future application to high-intensity, high-pressure combustion.

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